Summary
Details
Source
Context
Comments
Raw
Session
Compare
Tools
View
Export
GPU Throughput Chart
Memory Chart
Opcode Category Chart
% Occupancy Graphs

	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current	906 - flash_attention	(1, 32, 1)x(256, 1, 1)	856.38 us	898,176	0 - NVIDIA RTX A4500	1.05 Ghz	[23332] oa_out.exe	

GPU Speed Of Light Throughput

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	21.48	Duration [us]	856.38
Memory Throughput [%]	36.98	Elapsed Cycles [cycle]	898,176
L1/TEX Cache Throughput [%]	65.32	SM Active Cycles [cycle]	507,631.18
L2 Cache Throughput [%]	5.98	SM Frequency [Ghz]	1.05
DRAM Throughput [%]	0.76	DRAM Frequency [Ghz]	7.59

Small Grid This kernel grid is too small to fill the available resources on this device, resulting in only 0.6 full waves across all SMs. Look at ▸Launch Statistics for more details.

▸ Key Performance Indicators

Roofline Analysis The ratio of peak float (FP32) to double (FP64) performance on this device is 64:1. The workload achieved 5% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the ⊕Profiling Guide for more details on roofline analysis.

GPU Throughput
Compute (SM) [%]
Memory [%]
0.0
10.0
20.0
30.0
40.0
50.0
60.0
70.0
80.0
90.0
100.0
Speed Of Light (SOL) [%]

PM Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over its runtime.

Maximum Sampling Interval [us]	1	# Pass Groups	2
Maximum Buffer Size [Mbytes]	48	Dropped Samples [sample]	0

0s
ms
+0.1ms
+0.2ms
+0.3ms
+0.4ms
+0.5ms
+0.6ms
+0.7ms
+0.8ms
+0.9ms
+1ms
Overview
Average Active Warps Per Cycle
4.60 warp
0
Total Active Warps Per Cycle
257.35 warp
0
Blocks Launched
32 block
0
SM Active Cycles
605.93 cycle
0
Executed Ipc Active
..6 inst/cycle
0
SM
SM Throughput
100 %
0
SM ALU Pipe Throughput
100 %
0
SM FMA Light Pipe Throughput
100 %
0
SM FMA Heavy Pipe Throughpu
100 %
0
SM Tensor Pipe Throughput
100 %
0
SM XU Pipe Throughput
100 %
0
DRAM
DRAM Throughput
100 %
0
DRAM Read Bandwidth
100 %
0
DRAM Write Bandwidth
100 %
0
L2 Cache
L2 Throughput
100 %
0
L2 Hit Rate
100 %
0
L1 Cache
L1 Throughput
100 %
0
L1 Writeback Throughput
489.09 cycle
0
L1 Hit Rate
100 %
0
L1 Wavefronts (Data)
600.77
0
Workload Execution
flash_attention

Compute Workload Analysis

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	0.67	SM Busy [%]	29.80
Executed Ipc Active [inst/cycle]	1.19	Issue Slots Busy [%]	29.80
Issued Ipc Active [inst/cycle]	1.19		

Balanced Tensor is the highest-utilized pipeline (29.5%) based on active cycles, taking into account the rates of its different instructions. It is the logical aggregation of individual tensor pipelines. It's dominated by its Tensor (FP) sub-pipeline. It is well-utilized, but should not be a bottleneck.

Pipe Utilization (% of active cycles)
Tensor (All)
Tensor (FP)
FMA
ALU
FP64
Tensor (INT)
0.0
25.0
50.0
75.0
100.0
Utilization [%]
Pipe Utilization (% of peak instructions executed)
LSU
Tensor (FP)
FMA
ALU
XU
ADU
Uniform
CBU
FMA (FP16)
FP64 (DMMA)
FP64 (FP64)
TEX
Tensor (INT)
0.0
25.0
50.0
75.0
100.0
Utilization [%]

Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each memory unit.

Memory Throughput [Gbyte/s]	4.61	Mem Busy [%]	36.98
L1/TEX Hit Rate [%]	5.35	Max Bandwidth [%]	25.84
L2 Hit Rate [%]	137.85	Mem Pipes Busy [%]	21.48
L2 Compression Input Sectors [sector]	0	L2 Compression Success Rate [%]	0
L2 Compression Ratio	0	-	-

L1TEX Global Store Access Pattern
Est. Speedup: 27.74%
The memory access pattern for global stores to L1TEX might not be optimal. On average, only 8.0 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the ▸Source Counters section for uncoalesced global stores.

▸ Key Performance Indicators

Shared Load Bank Conflicts
Est. Speedup: 16.33%
The memory access pattern for shared loads might not be optimal and causes on average a 4.0 - way bank conflict across all 4194304 shared load requests.This results in 4194307 bank conflicts, which represent 25.00% of the overall 16777344 wavefronts for shared loads. Check the ▸Source Counters section for uncoalesced shared loads.

▸ Key Performance Indicators

Shared Store Bank Conflicts
Est. Speedup: 32.66%
The memory access pattern for shared stores might not be optimal and causes on average a 2.0 - way bank conflict across all 524288 shared store requests.This results in 524288 bank conflicts, which represent 50.00% of the overall 1048576 wavefronts for shared stores. Check the ▸Source Counters section for uncoalesced shared stores.

▸ Key Performance Indicators

Memory Chart
Values: Transfer Size
Inactivity: Greyed Out
Kernel
16.38 K Inst
Global
0.00 Req
16.38 K Req
0.00 Inst
Local
0.00 Req
0.00 Req
0.00 Inst
Texture
0.00 Req
0.00 Inst
Surface
0.00 Req
0.00 Req
133.12 K Inst
Load Global Store Shared
0.00 B
4.72 M Inst
Shared
4.19 M Req
524.29 K Req
Shared Memory
2.17 M Req
ICC Hit Rate: 99.94 %
232.19 K Req
IMC Hit Rate: 99.56 %
0.00 Req
IDC Hit Rate: 0.00 %
L1/TEX Cache Hit Rate: 5.35 %
68.16 MB
68.16 MB
4.19 MB
L2 Cache Hit Rate: 137.85 %
0.00 B
0.00 B
System Memory
3.19 MB
755.84 KB
Device Memory
0.00 B
0.00 B
Peer Memory
0.00 B
0.00 B
L2 Compression Ratio: 0.00
235.52 KB
9.52 K Req
512.00 Req
GCC Hit Rate: 97.45 %
% Peak
100%
80%
60%
40%
20%
0%

Scheduler Statistics

Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Issued Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates poor latency hiding.

Active Warps Per Scheduler [warp]	2.00	No Eligible [%]	70.20
Eligible Warps Per Scheduler [warp]	0.34	One or More Eligible [%]	29.80
Issued Warp Per Scheduler	0.30		

Issue Slot Utilization
Est. Local Speedup: 63.02%
Every scheduler is capable of issuing one instruction per cycle, but for this workload each scheduler only issues an instruction every 3.4 cycles. This might leave hardware resources underutilized and may lead to less optimal performance. Out of the maximum of 12 warps per scheduler, this workload allocates an average of 2.00 active warps per scheduler, but only an average of 0.34 warps were eligible per cycle. Eligible warps are the subset of active warps that are ready to issue their next instruction. Every cycle with no eligible warp results in no instruction being issued and the issue slot remains unused. To increase the number of eligible warps, avoid possible load imbalances due to highly different execution durations per warp. Reducing stalls indicated on the ▸Warp State Statistics and ▸Source Counters sections can help, too.

▸ Key Performance Indicators

Warps Per Scheduler
GPU Maximum Warps Per Scheduler
Theoretical Warps Per Scheduler
Active Warps Per Scheduler
Eligible Warps Per Scheduler
Issued Warp Per Scheduler
0.0
4.0
8.0
12.0
16.0

Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

Warp Cycles Per Issued Instruction [cycle]	6.71	Avg. Active Threads Per Warp	32
Warp Cycles Per Executed Instruction [cycle]	6.71	Avg. Not Predicated Off Threads Per Warp	31.82

Short Scoreboard Stalls
Est. Speedup: 48.93%
On average, each warp of this workload spends 3.3 cycles being stalled waiting for a scoreboard dependency on a MIO (memory input/output) operation (not to L1TEX). The primary reason for a high number of stalls due to short scoreboards is typically memory operations to shared memory. Other reasons include frequent execution of special math instructions (e.g. MUFU) or dynamic branching (e.g. BRX, JMX). Consult the Memory Workload Analysis section to verify if there are shared memory operations and reduce bank conflicts, if reported. Assigning frequently accessed values to variables can assist the compiler in using low-latency registers instead of direct memory accesses. This stall type represents about 48.9% of the total average of 6.7 cycles between issuing two instructions.

▸ Key Performance Indicators

Warp Stall Check the ▸Warp Stall Sampling (All Samples) table for the top stall locations in your source based on sampling data. The ⊕Profiling Guide provides more details on each stall reason.

Warp State (All Cycles)
Warp States
Stall Short Scoreboard
Stall Wait
Selected
Stall MIO Throttle
Stall Barrier
Stall Not Selected
Stall Math Pipe Throttle
Stall Long Scoreboard
Stall Dispatch Stall
Stall IMC Miss
Stall Branch Resolving
Stall No Instruction
Stall LG Throttle
Stall Drain
Stall Misc
Stall Membar
Stall Sleeping
Stall Tex Throttle
0.0
1.0
2.0
3.0
4.0
Cycles per Instruction

Instruction Statistics

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	33,879,040	Avg. Executed Instructions Per Scheduler [inst]	151,245.71
Issued Instructions [inst]	33,882,752	Avg. Issued Instructions Per Scheduler [inst]	151,262.29

FP32 Non-Fused Instructions
Est. Speedup: 3.83%
This kernel executes 6307840 fused and 7143936 non-fused FP32 instructions. By converting pairs of non-fused instructions to their ⊕fused, higher-throughput equivalent, the achieved FP32 performance could be increased by up to 27% (relative to its current performance).

▸ Key Performance Indicators

Executed Instruction Categories
Opcode Categories
Floating Point Instructions
Integer Instructions
Load/Store Instructions
Movement Instructions
Other
Miscellaneous Instructions
Control Instructions
Uniform Datapath Instructions
Conversion Instructions
0.0
5,000,000.0
10,000,000.0
15,000,000.0
20,000,000.0
Executed Warp-Level Instructions/Opcode

NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Topology

The system does not have any NVLink connections.

NVLink Tables

Detailed tables with properties for each NVLink.

Logical NVLink Properties

The system does not have any NVLink connections.

NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

NUMA ID Table

NUMA information is not available on the target system.

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	32	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	168	Static Shared Memory Per Block [byte/block]	0
Block Size	256	Dynamic Shared Memory Per Block [Kbyte/block]	73.73
Threads [thread]	8,192	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	0.57	Shared Memory Configuration Size [Kbyte]	102.40
Uses Green Context	0	Stack Size	1,024
# SMs [SM]	56	# TPCs	28
Enabled TPC IDs	all	-	-

Small Grid
Est. Speedup: 42.86%
The grid for this launch is configured to execute only 32 blocks, which is less than the GPU's 56 multiprocessors. This can underutilize some multiprocessors. If you do not intend to execute this kernel concurrently with other workloads, consider reducing the block size to have at least one block per multiprocessor or increase the size of the grid to fully utilize the available hardware resources. See the ⊕Hardware Model description for more details on launch configurations.

▸ Key Performance Indicators

Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	16.67	Block Limit Registers [block]	1
Theoretical Active Warps per SM [warp]	8	Block Limit Shared Mem [block]	1
Achieved Occupancy [%]	16.66	Block Limit Warps [block]	6
Achieved Active Warps Per SM [warp]	8.00	Block Limit SM [block]	16

Theoretical Occupancy
Est. Speedup: 63.02%
The 2.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (16.7%) is limited by the number of required registers, and the required amount of shared memory.

▸ Key Performance Indicators

Impact of Varying Register Count Per Thread
% Occupancy
125
100
75
50
25
0
0
8
16
24
32
40
48
56
64
72
80
88
96
104
112
120
128
136
144
152
160
168
176
184
192
200
208
216
224
232
240
248
256
Registers Per Thread
Impact of Varying Block Size
% Occupancy
125
100
75
50
25
0
0
32
64
96
128
160
192
224
256
288
320
352
384
416
448
480
512
544
576
608
640
672
704
736
768
800
832
864
896
928
960
992
1,024
Block Size
Impact of Varying Shared Memory Usage Per Block
% Occupancy
125
100
75
50
25
0
0
3,184
6,368
9,552
12,736
15,920
19,104
22,288
25,472
28,656
31,840
35,024
38,208
41,392
44,576
47,760
50,944
54,128
57,312
60,496
63,680
66,864
70,048
73,232
76,416
79,600
82,784
85,968
89,152
92,336
95,520
98,704
101,888
Shared Memory Per Block

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	507,631.18	Average L1 Active Cycles [cycle]	507,631.18
Average L2 Active Cycles [cycle]	177,600.80	Average SMSP Active Cycles [cycle]	507,582.06
Average DRAM Active Cycles [cycle]	49,353.60	Total SM Elapsed Cycles [cycle]	50,214,792
Total L1 Elapsed Cycles [cycle]	50,214,792	Total L2 Elapsed Cycles [cycle]	38,022,800
Total SMSP Elapsed Cycles [cycle]	200,859,168	Total DRAM Elapsed Cycles [cycle]	65,026,048

SMs Workload Imbalance
Est. Speedup: 24.50%
One or more SMs have a much lower number of active cycles than the average number of active cycles. Maximum instance value is 43.27% above the average, while the minimum instance value is 100.00% below the average.

▸ Key Performance Indicators

SMSPs Workload Imbalance
Est. Speedup: 24.49%
One or more SMSPs have a much lower number of active cycles than the average number of active cycles. Maximum instance value is 43.26% above the average, while the minimum instance value is 100.00% below the average.

▸ Key Performance Indicators

L1 Slices Workload Imbalance
Est. Speedup: 24.50%
One or more L1 Slices have a much lower number of active cycles than the average number of active cycles. Maximum instance value is 43.27% above the average, while the minimum instance value is 100.00% below the average.

▸ Key Performance Indicators

L2 Slices Workload Imbalance
Est. Speedup: 5.63%
One or more L2 Slices have a much higher number of active cycles than the average number of active cycles. Maximum instance value is 30.13% above the average, while the minimum instance value is 2.56% below the average.

▸ Key Performance Indicators

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	507,631.18	0	894,892	28,427,346
SMSP Active Cycles	507,582.06	0	894,590	113,698,382
L1 Active Cycles	507,631.18	0	894,892	28,427,346
L2 Active Cycles	177,600.80	173,053	254,187	7,104,032
DRAM Active Cycles	49,353.60	48,880	49,872	493,536

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	164,096	Branch Efficiency [%]	100
Branch Instructions Ratio [%]	0.00	Avg. Divergent Branches	0

Uncoalesced Global Accesses
Est. Speedup: 0.81%
This kernel has uncoalesced global accesses resulting in a total of 98304 excessive sectors (4% of the total 2260992 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The ⊕CUDA Programming Guide has additional information on reducing uncoalesced device memory accesses.

▸ Key Performance Indicators

L2 Theoretical Sectors Global Excessive

Location	Value	Value (%)
0xb00c769e0 in flash_attention	1,536	2
0xb00c76900 in flash_attention	1,536	2
0xb00c767e0 in flash_attention	1,536	2
0xb00c766c0 in flash_attention	1,536	2
0xb00c765a0 in flash_attention	1,536	2

Uncoalesced Shared Accesses
Est. Speedup: 14.55%
This kernel has uncoalesced shared accesses resulting in a total of 4718592 excessive wavefronts (26% of the total 18358272 wavefronts). Check the L1 Wavefronts Shared Excessive table for the primary source locations. The ⊕CUDA Best Practices Guide has an example on optimizing shared memory accesses.

▸ Key Performance Indicators

L1 Wavefronts Shared Excessive

Location	Value	Value (%)
0xb00c71fc0 in flash_attention	131,072	3
0xb00c71f50 in flash_attention	131,072	3
0xb00c71ee0 in flash_attention	131,072	3
0xb00c71e70 in flash_attention	131,072	3
0xb00c71e00 in flash_attention	131,072	3

Warp Stall Sampling (All Samples)

Location	Value	Value (%)
0xb00c72080 in flash_attention	570	3
0xb00c71730 in flash_attention	474	3
0xb00c719d0 in flash_attention	473	3
0xb00c71570 in flash_attention	468	3
0xb00c71e30 in flash_attention	463	3

Most Instructions Executed

Location	Value	Value (%)
0xb00c72080 in flash_attention	32,768	0
0xb00c72070 in flash_attention	32,768	0
0xb00c72060 in flash_attention	32,768	0
0xb00c72050 in flash_attention	32,768	0
0xb00c72010 in flash_attention	32,768	0

Follow the *rules outputs* to get guidance on how to navigate through the report and quickly discover performance bottlenecks in this kernel.
You could also disable individual sections to focus on selected performance aspects and make profiling faster.